

EAZYAI INTELLIGENCE

ARCHITECTURAL WHITEPAPER V1.0

Autonomous Talent Intelligence & Multimodal AI Interviewing

Complete end-to-end technical specification: Zero-trust BFF orchestration, distributed microservices mesh, multimodal proctoring, acoustic deepfake detection, vector RAG search, and OKLCH frosted glassmorphism.

CORE FRAMEWORK

TanStack Start + React 19

SERVER ENGINE

Nitro ESM + Google Cloud Run

SECURITY ARCHITECTURE

IAM Zero-Trust + HttpOnly Cookie

CLASSIFICATION

Technical Platform Specification

EXECUTIVE BRIEF

1. Problem Statement & Platform Vision

Legacy recruitment infrastructures suffer from catastrophic failure modes: manual resume screening fatigue, inability to discern AI-hallucinated candidate CVs, vulnerability to proxy test takers, and subjective, uncalibrated interview scoring. EazyAI replaces disjointed tooling with an autonomous talent mesh unifying semantic vector indexing, conversational virtual agents, and biometric anti-fraud proctoring into a single portal.

1. Semantic Vector RAG

Autonomous extraction, high-dimensional vector embeddings, and cosine similarity matching of candidate resumes against job requisition criteria with explicit gap detection.

2. Multimodal AI Interviews

Autonomous conversational interviews powered by Dialogflow CX, featuring real-time speech-to-text, acoustic waveform playback, video streaming, and technical rubric scoring.

3. Anti-Fraud Proctoring

Synthetic voice deepfake detection, computer vision webcam face verification, and gaze/window deviation tracking synthesized into a composite Trust Gauge (0-100%).

INDEX

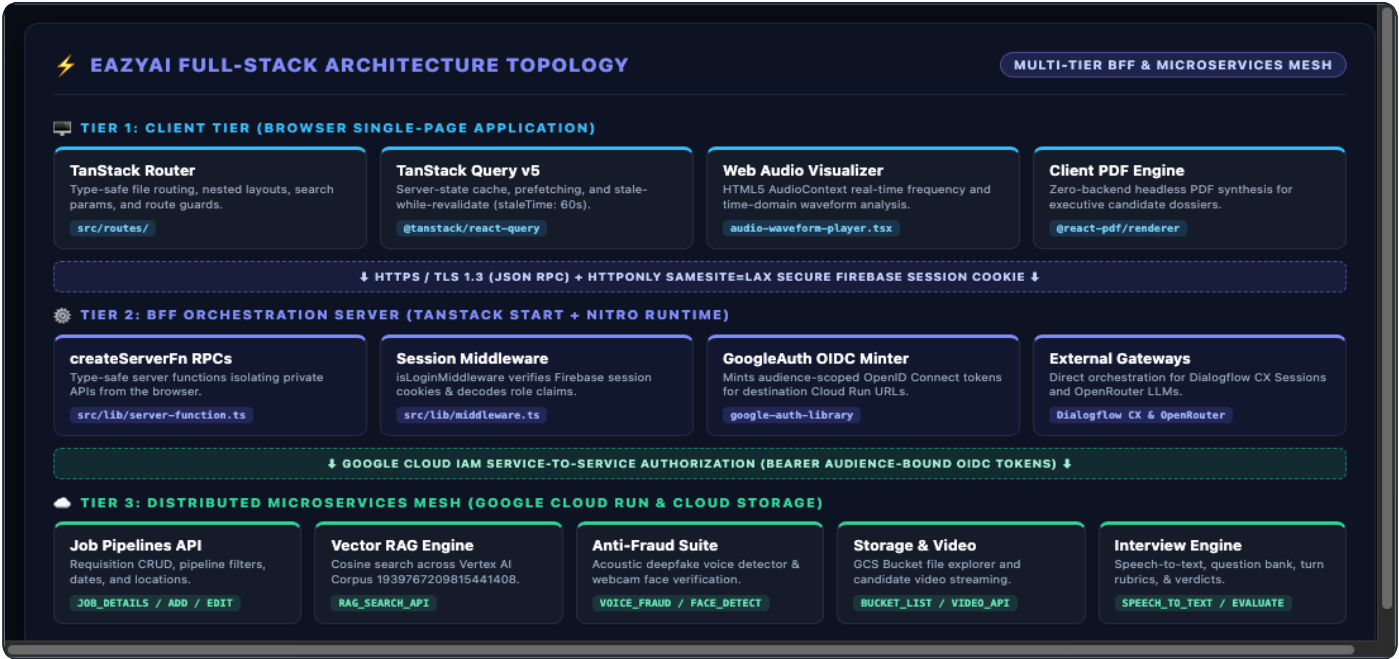
Document Navigation Table

PAGE	SECTION & MODULE	KEY ARCHITECTURAL ELEMENTS
Page 2	System Architecture & Architectural Diagram	BFF Pattern, Nitro Engine, Full Visual System Diagram
Page 3	Zero-Trust Security & Auth Flow Diagram	Firebase Admin Cookies, Google IAM OIDC Minting, RBAC Matrix
Page 4	Microservices API Specifications (Part 1)	Jobs, Ingestion, Vector RAG Search & Storage Endpoints
Page 5	Microservices API Specs (Part 2) & External AI	Dialogflow CX Sessions, OpenRouter LLM, Evaluation & Video
Page 6	Candidate Ingestion, Archive & Vector RAG	GCS Explorer, Corpus Indexing, Semantic Match Algorithm
Page 7	Question Bank, AI Generation & Simulator	Rubric Calibrator, OpenRouter Prompts, Interactive Sandbox
Page 8	Multimodal Evaluation & Session Timeline	Web Audio Visualizer, Video Streams, Dialogflow CX Turns
Page 9	Anti-Fraud Suite & Trust Gauge Diagram	Deepfake Voice Detector, Face Cosine Match, Trust Formula
Page 10	Evaluator Workbench Copilot & PDF Dossiers	LocalStorage Scratchpad, Weighted Sliders, Email Inbound Sync
Page 11	Governance & Glassmorphism Design System	OKLCH Palette, Glass Tokens, Typography, K Command Palette
Page 12	DevOps, Deployment & Infrastructure Guide	Multi-Stage Dockerfile, Firebase App Hosting, Cloud Run

ARCHITECTURE BLUEPRINT

2. High-Level System Architecture

EazyAI is architected around a Backend-For-Frontend (BFF) topology powered by TanStack Start and the Nitro server runtime. The diagram below illustrates the end-to-end component layers, network boundaries, and communication protocols.



CORE RATIONALE

BFF Architectural Design Principles

Why Backend-For-Frontend (BFF)?

- **Eliminates Client Secret Leakage:** Cloud Run microservice URLs, Firebase Admin credentials, and OpenRouter API keys never touch the browser DOM or network bundles.
- **Enforces IAM Zero-Trust:** Cloud Run microservices remain private (*-ingress: internal-and-cloud-load-balancing*). Direct internet invocations are blocked with HTTP 403.
- **Prevents CORS Proliferation:** The BFF aggregates responses from 20+ microservices on the server side, presenting a unified, typed API to the client.

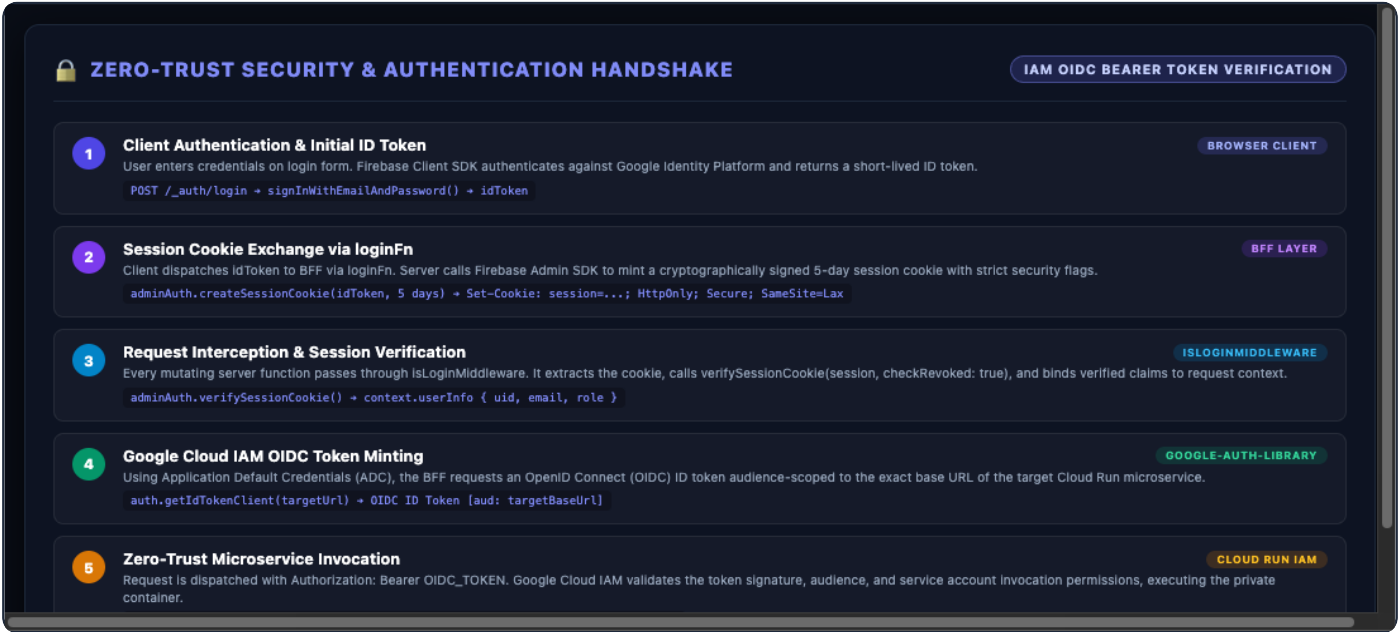
Nitro Server Engine & Hydration

- **High-Performance ESM Bundle:** Nitro compiles the server application to a lean standalone ESM bundle (`.output/server/index.mjs`) executing on Node 20.
- **Zero Layout Shift Hydration:** Route loaders execute server queries via `ensureQueryData`, transmitting dehydrated HTML directly to the browser.
- **Native Cookie Streaming:** Utilizes `@tanstack/react-start/server` cookie helpers (`getCookie`, `setCookie`) for seamless header injection.

SECURITY BLUEPRINT

3. Zero-Trust Security & Authentication Protocol

EazyAI establishes an airtight security perimeter. Authentication is decoupled into two isolated domains: (1) Client-to-BFF session management backed by Firebase Admin HTTP-only cookies, and (2) BFF-to-Cloud-Run machine-to-machine authentication enforcing Google Cloud IAM OIDC bearer token verification.



RBAC GOVERNANCE

Role-Based Access Control (RBAC) Matrix

ROLE TITLE	SCOPE OF AUTHORITY	PERMITTED ACTIONS	RESTRICTED OPERATIONS
Super Admin	Platform Governance	User role assignment, account restriction, site config, vector corpus triggers, full hiring operations	None (Full Administrative Control)
Recruiter	Talent & Job Pipeline	Job creation/editing, resume uploads, vector RAG search, candidate comparison, scheduling	Role modification, system configs
Interviewer	Candidate Evaluation	Answer review, audio playback, video review, scratchpad notes, submit verdict (ACCEPT/REJECT)	Job requisition deletion, admin panels
Viewer / Auditor	Read-Only Inspection	Inspect dashboard metrics, view candidate profiles, read-only interview review, download PDF reports	All mutating operations, note edits

Session Attack Surface Mitigation

Session cookies are protected by **HttpOnly: true** (inaccessible to malicious scripts), **SameSite: 'lax'** (blocks cross-site request forgery), and **Secure: true** in production environments. Tokens are verified on every invocation.

Audience Scoping & Zero Token Replay

Every machine token generated by *google-auth-library* is audience-restricted to the destination service URL. An intercepted token destined for the Job API cannot be replayed against the Anti-Fraud or RAG APIs.

API CATALOG

4. Microservices Catalog: Requisitions, Ingestion & Vector RAG

Downstream microservice endpoints are declared in `src/lib/api-path.ts`. The table below specifies input payloads, response contracts, regions, and associated BFF server functions.

SERVICE KEY / PATH	REGION	INPUT SCHEMA / PARAMETERS	OUTPUT DATA CONTRACT	SERVER FUNCTION
JOB_DETAILS <code>/api/jobs-list-api</code>	us-central1	limit?: number, cursor?: string, status?: 'Active' 'Inactive'	PaginatedJobResponse: count, next_cursor, data: JobDetail[]	getJobDetails
ADD_JOB <code>/api/jobs</code>	us-central1	jobTitle, jobDescription, jobType, experience, locations[], dates	JobPosting: id, status, createdAt	createJob
EDIT_JOB <code>/api/jobs</code>	us-central1	jobId, jobTitle, jobDescription, locations[], status, experience	JobPosting: updated metadata record	editJob
CANDIDATE_LIST <code>/api/candidates</code>	us-central1	limit?: number, cursor?: string, search?: string	PaginatedCandidateResponse: count, next_cursor, candidate[]	getCandidatesList
ADD_CANDIDATE <code>/api/upload</code>	us-central1	FormData: file (PDF/DOCX), candidate_name, candidate_email	UploadResponse: id, resume_url, uploaded_at, candidate_id	addCandidate
ADD_MULTIPLE_CANDIDATES <code>/api/v1/jobs/upload-cvs</code>	us-central1	FormData: files[] (batch multipart), job_id: string	MultipleCvUploadResponse: uploaded_files[], failed_files[]	addMultipleCandidates
BUCKET_LIST_API <code>/api/list-objects</code>	us-central1	prefix: 'uploads', pageToken?: string	BucketListResponse: root_folders[], root_files[], next_page_token	fetchBucketListInfo
TRIGGER_INDEX <code>/api/trigger-indexing</code>	us-central1	corpus_id: '1939767209815441408', selected_paths[]	IndexJobResponse: job_id, status: 'TRIGGERED', timestamp	triggerIndexes
RAG_SEARCH_API <code>/api/rag_search</code>	us-central1	jobId, jobDescription, experience, skills, domain, file_ids[]	ProfileSearchResponse: matches: CandidateMatch[] (score, gaps)	getSearchProfileDetails

Signed URL Generation for Media Assets
The `DOWNLOAD_FILE_URL` and `DOWNLOAD_VOICE_FILE_URL` services generate time-limited, signed Google Cloud Storage URLs. Resume PDFs and voice MP3s are accessed securely without exposing public bucket read permissions.

Indexed Files State Tracking
The `PROCESSED_FILES_ID` endpoint delivers the complete list of document IDs persisted in the vector store. This powers the Archive Bank explorer, displaying real-time indexing status badges on resumes.

API CATALOG

5. Evaluation, Anti-Fraud, Video & External AI Gateways

SERVICE KEY / PATH	REGION	INPUT SCHEMA / PARAMETERS	OUTPUT DATA CONTRACT	SERVER FUNCTION
SPEECH_TO_TEXT /api/transcribe-v2	us-central1	audio_gcs_uri: string, candidate_email: string	TranscribeResponse: text: string, confidence: number	transcribeCandidateAudio
VOICE_FRAUD_DETECTION /api/interview/audio-analysis	us-central1	candidateEmail: string, jobId: string	AudioAnalysisResponse: overall_conclusion, confidence, records[]	getAudioAnalysisResultFn
FACE_DETECTION /api/match	us-central1	webcam_base64: string, resume_photo_url: string	FaceMatchResponse: is_match: boolean, confidence: number	verifyFaceRecognition
MOVEMENT_OUTCOME /api/movement-detection	us-central1	candidate_email: string, job_id: string	MovementOutcomeResponse: total_events, events: UserEvent[]	getMovementDetectionDetails
INTERVIEW_VIDEO_API /api/interview-video	us-central1	candidate_email: string, session_id?: string	InterviewVideoResponse: count, data: InterviewVideoRecord[]	getInterviewVideoList
INTERVIEW_EVALUTE /api/evaluate	us-central1	candidate_email, job_id, verdict: 'ACCEPT' 'REJECT', feedback	EvaluateResponse: status: 'SUCCESS', record_id	interviewEvaluate
ADMIN_USER_LIST / ACT /api/admin-users-list	us-central1	target_uid, role: string, disabled: boolean	AdminUserResponse: count, data: UserData[]	adminUsersList / adminActivity

AI GATEWAYS

External AI Gateways: Dialogflow CX & OpenRouter SDK

Dialogflow CX Runtime Protocol

The virtual interviewer is powered by `@google-cloud/dialogflow-cx`. The BFF calls `client.projectLocationAgentSessionPath()` to bind sessions to Project `project-716b1c69-ee04-40fd-ba6` and Agent `01a7a4d2-bdec-43b9-9ab2-be8486843872`. Transcribed candidate audio is processed via `detectIntent()`, returning audio MP3 streams and intent parameters.

OpenRouter LLM Synthesis

Integrates with `@openrouter/sdk` using high-capacity models like `nvidia/nemotron-3-super-120b-a12b:free`. The BFF formats role prompts in `getJobDescription` and question generation modules, receiving calibrated job requisitions and evaluation rubrics without managing GPU infrastructure.

ARCHIVE & VECTOR SEARCH

6. Resume Ingestion, Cloud Storage Bank & Vector RAG

EazyAI integrates Google Cloud Storage bucket exploration with high-dimensional vector embeddings, enabling recruiters to ingest thousands of resumes, index them into a specialized corpus, and execute multi-dimensional semantic queries.

Archive Bank Explorer (/dashboard/import)

Provides an in-browser virtual file system directly querying Google Cloud Storage prefixes. Recruiters inspect folder trees, review document sizes and MIME types, download signed URLs, and select file subsets to trigger asynchronous vector re-indexing into Corpus **1939767209815441408**.

High-Throughput Batch CV Ingestion

The drag-and-drop batch upload modal (*add-multiple-candidates-dialog.tsx*) leverages multipart streaming (*ADD_MULTIPLE_CANDIDATES*) to upload dozens of CVs concurrently. The pipeline parses applicant names, emails, and phone numbers, linking them to requisition IDs with real-time progress indicators.

VECTOR DISCOVERY

Semantic RAG Discovery Engine & Gap Analysis

SCORING DIMENSION	VECTOR MECHANICS & ALGORITHMIC PROCESS	RECRUITER ACTIONABLE OUTPUT
Match Score (0-100%)	Cosine similarity computed between requisition query embedding and indexed resume text chunk embeddings.	Color-coded match badge for immediate candidate ranking.
Seniority Inference	Heuristic deduction assessing total professional years, team leadership keywords, and architectural responsibilities.	Categorized as Junior, Mid, Senior, or Lead architect.
Matched Criteria	Positive vector projection identifying explicit qualifications, frameworks, and domain expertise.	List of explicitly satisfied criteria with source citations.
Missing Information	Negative constraint analysis identifying requisition requirements omitted or unverified in the candidate resume.	Flags skill gaps for focused questioning during interviews.

Candidate Comparison Dialog

Side-by-side comparison modal (*candidate-compare-dialog.tsx*) allowing recruiters to evaluate shortlisted candidates across match scores, experience levels, interview verdicts, and biometric trust gauges.

Competency Radar Visualization

The Skill Radar (*candidate-skill-radar.tsx*) renders a Recharts polar chart depicting candidate mastery across core technical competencies versus target job benchmark requirements.

QUESTION CURATION

7. Requisition Question Bank & AI Question Synthesis

EazyAI combines automated GenAI question formulation with manual recruiter governance to ensure assessment interviews are calibrated to requisition requirements and standardized across all candidates.

Requisition-Specific Question Repository

Managed via `/dashboard/questions`. Questions are mapped directly to job IDs and categorized across technical coding, system design, architectural trade-offs, and behavioral communication. Recruiters can manually add, edit, or delete questions via `QUESTION_ADD` and `QUESTION_DELETE`.

GenAI Question Synthesis (`QUESTION_ADD_AI`)

Invokes an LLM microservice that parses the requisition title, description, and required tech stack. Synthesizes structured questions with evaluation rubrics, key terminology checklists, and expected answer criteria, ensuring interviewers have objective grading benchmarks.

SIMULATION SANDBOX

Interactive Candidate Experience Simulator

The Question Simulator (`question-simulator-dialog.tsx`) allows hiring teams to step into the candidate shoes and test the automated interview flow before publishing questions live.

SIMULATION STAGE	SANDBOX EMULATION BEHAVIOR	QUALITY ASSURANCE OBJECTIVE
1. Introduction Turn	Virtual avatar introduces position, states rules, and prompts for candidate self-introduction.	Verify conversational clarity and pacing.
2. Question Pacing	Delivers technical questions with countdown clocks, thinking time, and answer recording limits.	Calibrate time limits for complex design questions.
3. Speech Transcription	Emulates microphone input and displays simulated real-time speech-to-text transcript output.	Check technical acronym transcription accuracy.
4. Rubric Grading Preview	Shows immediate rubric comparison against simulated answer to preview scoring behavior.	Ensure rubric thresholds are realistic and fair.

Simulator Sandbox Isolation: The question simulator operates in complete isolation from the production candidate database. Test audio and transcripts generated during simulation sessions are flagged as test telemetry and never pollute candidate archives.

MULTIMODAL REVIEW

8. Multimodal Interview Outcomes & Audio Player

The Interview Outcome dashboard (*/dashboard/interview/\$id*) unifies spoken transcripts, acoustic waveform players, video playback, and turn-by-turn conversational telemetry into an all-in-one recruiter console.

Answer Outcome & Rubric Inspector

The Answer Outcome component (*answer-outcome.tsx*) displays candidate spoken answers aligned with question prompts. AI evaluation cards break down: (1) AI Verdict, (2) Domain competency score (0-100), (3) Structured technical reasoning, and (4) Keyword rubric inspector.

Web Audio API Waveform Visualizer

The Audio Waveform player (*audio-waveform-player.tsx*) initializes an in-browser **AudioContext** with real-time frequency-domain and time-domain canvas visualizers. Supports variable playback rates (0.75x, 1.0x, 1.25x, 1.5x, 2.0x) and interactive scrubber seeking.

VIDEO STREAMING

Interview Video Streaming & Outcome Preview

The Video Outcome component (*video-outcome.tsx*) streams interview video recordings directly from Google Cloud Storage via signed URLs (*INTERVIEW_VIDEO_API*):

VIDEO STREAM METRIC	COMPONENT CAPABILITY	RECRUITER INSPECTION VALUE
Stream Playback	HTML5 Video with full transport controls, buffering indicators, and duration markers.	Inspect non-verbal communication and physical demeanor.
Metadata Tracking	File size formatting (e.g. 42.5 MB), MIME type (video/mp4), and session timestamps.	Verifies video integrity and completeness.
Time Synchronization	Synchronized timestamp markers matching question prompt intervals.	Jump directly to specific question video segments.

TELEMETRY

Conversational Session Timeline & Token Telemetry

Turn-by-Turn Conversational Events

The Session Timeline (*session-timeline.tsx*) reconstructs the complete Dialogflow CX conversational journey. Evaluators can inspect each node invocation, intent matched, thought signatures, and tool execution parameters.

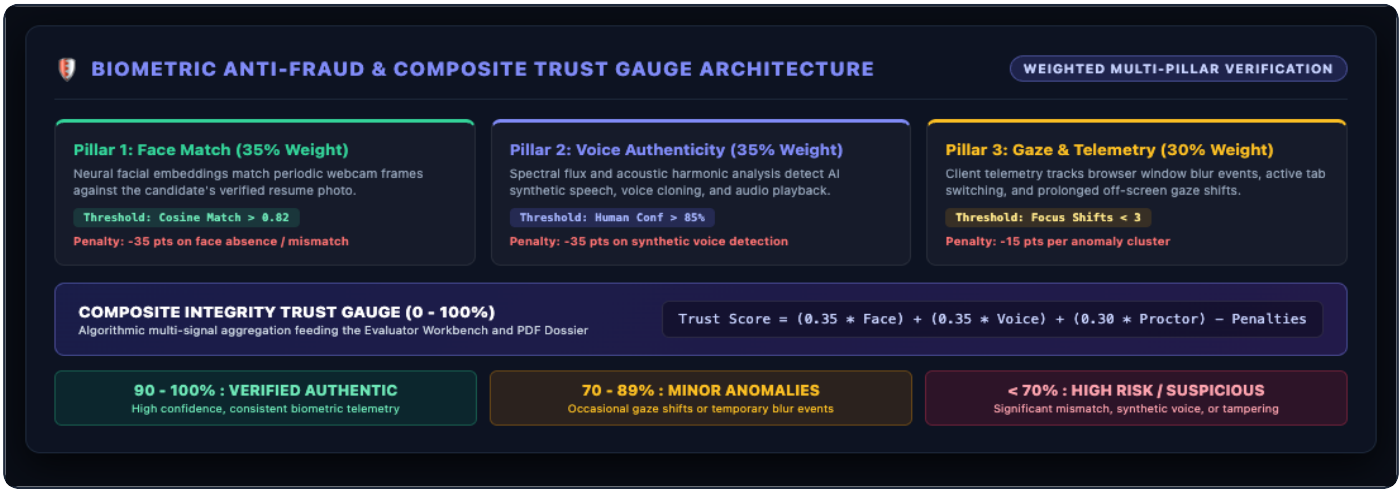
Token Consumption Breakdown

Logs detailed token usage: Prompt Token Count, Candidate Token Count, Total Tokens Consumed, and modality splits (Text vs. Audio tokens). Ensures cost auditability across every candidate interview session.

BIOMETRIC SECURITY

9. Anti-Fraud Proctoring & Trust Gauge Architecture

Remote recruitment is increasingly vulnerable to AI-assisted cheating, proxy test takers, and synthetic speech cloning. EazyAI combines acoustic waveform forensics, computer vision face verification, and proctoring telemetry into a unified trust score.



BIOMETRIC TELEMETRY

Anti-Fraud Verification Dimensions

Face Anti-Impersonation
Compares webcam captures taken during the interview against candidate resume photos using neural face embeddings (*FACE_DETECTION*). Detects candidate substitution or proxy attendance during live questions.

Acoustic Deepfake Detection
The Audio Analysis engine (-*VOICE_FRAUD_DETECTION*) inspects acoustic spectral flux, cloning jitter, and synthetic artifacts, classifying recordings as **Human**, **Suspicious**, or **AI-Generated**.

Movement & Gaze Proctoring
Telemetry tracking logs browser tab switches, background application focus changes, out-of-frame head deviations, and erratic gaze shifts during active questioning (*MOVEMENT_OUTCOME*).

ANOMALY TYPE	DETECTION MECHANISM	TRUST PENALTY	FLAG SEVERITY
Face Mismatch	Cosine distance > threshold	-35 Points	CRITICAL (Manual Review)
Voice Cloning Detected	Synthetic audio probability > 80%	-35 Points	CRITICAL (Fraud Flag)
Multiple Window Focus Loss	Browser blur event count > 4	-15 Points	WARNING (Proctor Alert)
Continuous Gaze Deviation	Off-screen gaze for > 12 seconds	-10 Points	WARNING (Cheating Risk)

EVALUATOR COPILOT

10. Evaluator Workbench Copilot & PDF Dossiers

EazyAI equips interviewers with a client-side copilot environment for conducting synchronized, evidence-based evaluations with persistent scratchpad notes, weighted composite scoring sliders, and instant PDF dossier exports.

Evaluator Scratchpad Copilot

The Scratchpad (*evaluator-scratchpad.tsx*) automatically persists evaluator notes in **localStorage** keyed by candidate and job ID. Evaluators can jot observations during review without losing state on accidental browser reloads or tab closures.

Weighted Composite Scoring & Quick Verdicts

Includes dynamic weight sliders across Technical Competency, Communication, and Cultural Fit. Offers one-click verdict templates ("Strong Hire", "Solid Contender", "Reject - Technical Gap", "Reject - Integrity Flag") to accelerate hiring committee decisions.

DOSSIER ENGINE

Client-Side Executive Candidate Dossier Export

Powered by `@react-pdf/renderer`, the platform dynamically renders multi-page PDF candidate evaluation dossiers directly in the browser with zero server CPU overhead.

Dossier Section	Synthesized Data Elements	Hiring Committee Value
Executive Cover	Candidate demographics, target job, interview date, composite score badge, and final verdict.	Instant C-suite executive briefing.
Trust & Integrity Gauge	Visual Trust score, face match confidence, acoustic voice authenticity, proctoring anomalies.	Guarantees candidate authenticity.
Question Breakdown	Question prompts, spoken answer transcripts, domain scores, and AI rubric rationale.	Auditable evaluation trail.
Recruiter Notes Block	Persistent scratchpad observations, interviewer feedback, compensation notes, and signature.	Formal recruitment record.

AUTOMATION

Inbound Email Application Synchronization

Automated Inbox Polling

The Email Sync microservice (*EMAIL_SYNC*) ingests candidate application emails sent to corporate hiring mailboxes, parsing applicant contact info, email text bodies, and attached resume files.

Auto-Requisition Mapping

The Email Reader dialog (*email-reader-dialog.tsx*) allows recruiters to inspect the raw email, preview the attached resume, and confirm automated association with active Job Requisition IDs.

DESIGN SYSTEM

11. Frosted Glassmorphism Design System (OKLCH)

EazyAI is engineered to deliver a state-of-the-art visual experience. Built on **Tailwind CSS v4**, the interface employs the **OKLCH** color space for vibrant, perceptually uniform tones that interact naturally with semi-transparent frosted glass layers.

Design Utility Token	Tailwind V4 Styling Rule	OKLCH Light / Dark	UI Placement Role
glass-header	bg-background/50, backdrop-blur-2xl	Light: 50% / Dark: 70%	Sticky top navigation bar
glass-sidebar	bg-sidebar/40, backdrop-blur-2xl	Light: 40% / Dark: 40%	Primary app navigation panel
glass-card	bg-card/30, backdrop-blur-lg, shadow-lg	Light: 30% / Dark: 30%	Metric widgets, data tables, dialogs
ambient-pulse	15s breathing pulse keyframe	Indigo / Violet glow	Dual corner ambient background

TYPOGRAPHY

Typography Scale & Font Foundations

Outfit Variable

Used for page heroes, section titles, and numerical stat cards. Delivers a high-contrast modern SaaS brand presence with tight letter tracking.

Geist Sans Variable

The primary sans-serif workhorse font applied to body text, form controls, table cells, and button labels.

Geist Mono Variable

Monospace font applied to timestamps, session IDs, token metrics, and technical API payloads.

PRODUCTIVITY

Command Palette (K) & Guided Onboarding Tour

Global Search Palette (K / Ctrl+K)

The Command Palette (*global-search-dialog.tsx*) provides instant keyboard navigation across all dashboard routes, active jobs, candidate profiles, and admin tools with fuzzy search filtering.

Interactive Onboarding Walkthrough

The Feature Tour dialog (*onboarding-tour-dialog.tsx*) guides new recruiters through the core workflows on first login. Progress is automatically cached in **localStorage**.

DEVOPS & DEPLOYMENT

12. Production Deployment, Docker & Firebase App Hosting

EazyAI is designed for cloud-native deployment with zero downtime. The application container compiles into a standalone Nitro ESM server, optimized for Firebase App Hosting and Google Cloud Run.

DEPLOYMENT TARGET	CONFIGURATION & BUILD PIPELINE	OPERATIONAL & SCALING CHARACTERISTICS
Firestore App Hosting	Configured via apphosting.yaml. Automated builds with pnpm.Autoscaling: min 0, max 10 instances, concurrency: 80. Node Secrets mapped via Google Cloud Secret Manager.	options: --max-old-space-size=4096.
Google Cloud Run	Multi-stage Dockerfile packaging Node 20 slim image runningServerless auto-scaling per request volume. Native Google Cloud .output/server/index.mjs on port 8080.	IAM service account authentication.
Local Development	Vite dev server (pnpm dev) with HMR, TanStack Router Devtools, and React Query Devtools on port 3000.	Local authentication via gcloud auth application-default login for seamless Cloud Run proxying.

CONTAINER ARCHITECTURE

Multi-Stage Docker Container Architecture

Stage 1: Builder

Installs full devDependencies, copies source files, passes build-time client Firestore config args, and compiles the Nitro application via `pnpm build`.

Stage 2: Prod-Deps

Executes `pnpm install --prod --frozen-lockfile`, pruning devDependencies and build tools to produce a minimal, secure node_modules footprint.

Stage 3: Runner

Copies the compiled `.output` bundle and production node_modules into a clean Node 20 slim runtime. Exposes port 8080 and executes via `node .output/server/index.mjs`.

QUALITY ASSURANCE

Testing, Linting & CI/CD Verification

Automated Unit Testing with Vitest

Run via `pnpm test`. Validates components, form validation schemas, and media helpers with mocked browser APIs (HTMLMediaElement, AudioContext).

Linting & Formatting Standard

Standardized via TanStack ESLint rules and Prettier (`pnpm check`). Enforces strict TypeScript type checking, import sorting, and formatting rules across all modules.

PDF Specification Generation: This document is generated programmatically using `@react-pdf/renderer` via `pnpm docs:pdf`. Any updates to API endpoints, architectural topologies, or design tokens can be instantly compiled into a fresh PDF specification.